Lab 3: Descriptive Statistics

Code

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A. Weather Forecast Data

1. For this lab, we'll be using data on weather forecasts gathered by student at Saint Louis University. You can read about the dataset here. Download the weather forecasts data using the following code:

```
setwd("C:/Users/bdish/OneDrive/Documents/Math167R/data")
weather_forecasts <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesda
write.csv(weather_forecasts, "weather_forecasts.csv")
```

2. How many rows are in this dataset? How many columns?

```
nrow(weather_forecasts)
```

```
[1] 651968
```

```
ncol(weather_forecasts)
```

```
[1] 10
```

3. How many cities are represented in this dataset?

```
cleaned_cities <- unique(weather_forecasts$city, na.rm = T)
length(cleaned_cities)
```

```
[1] 160
```

4. Create a new data frame containing only the forecasts for San Jose. You may have to explore the values for the `city` variable.

```
library(dplyr)
sj_fcs <- weather_forecasts |>
  filter(city == "SAN_JOSE") |>
  select(forecast_hours_before, forecast_temp,
  forecast_outlook)
df_sj <- as.data.frame(sj_fcs, na.rm = T)
head(df_sj, n = 20)
```

```
   forecast_hours_before forecast_temp forecast_outlook
1                     48            NA             <NA>
```

```
2                    36            NA             <NA>
3                    24            NA             <NA>
4                    12            59           PTCLDY
5                    48            NA             <NA>
6                    36            NA             <NA>
7                    24            43           PTCLDY
8                    12            43           PTCLDY
9                    48            NA             <NA>
10                   36            63           MOCLDY
11                   24            63           PTCLDY
12                   12            64           PTCLDY
13                   48            46           MOCLDY
14                   36            46             RAIN
15                   24            45           PTCLDY
16                   12            45           PTCLDY
17                   48            63             RAIN
18                   36            63           PTCLDY
19                   24            64           PTCLDY
20                   12            64           PTCLDY
```

5. Compute the mean absolute error between `observed_temp` and `forecast_temp` for San Jose.

```r
library(dplyr)

sj_ot_ft <- weather_forecasts |>
  filter(city == "SAN_JOSE") |>
  mutate(abs_error = abs(observed_temp - forecast_temp))

mean(sj_ot_ft$abs_error, na.rm = T)
```

```
[1] 2.169762
```

6. Compute the mean absolute error between `observed_temp` and `forecast_temp` for San Jose using only forecasts made 48 hours in advance.

```r
# YOUR CODE HERE
library(dplyr)
sj_ot_ft_48 <- weather_forecasts |>
  filter(city == "SAN_JOSE", forecast_hours_before == 48) |>
  mutate(abs_error_48 = abs(observed_temp - forecast_temp))

mean(sj_ot_ft_48$abs_error_48, na.rm = T)
```

```
[1] 2.262544
```

7. Compute the mean absolute error between `observed_temp` and `forecast_temp` for San Jose using only forecasts made 12 hours in advance.

```r
library(dplyr)
sj_ot_ft_12 <- weather_forecasts |>
```

```
  filter(city == "SAN_JOSE", forecast_hours_before == 12) |>
  mutate(abs_error_12 = abs(observed_temp - forecast_temp))

mean(sj_ot_ft_12$abs_error_12, na.rm = T)
```

```
[1] 2.0553
```

8. Compare your answers to 6 and 7. What do you notice? How does this compare to your expectation?

Comparing answers for 6 and 7, we can see that the means abs. error for San Jose using forecasts made 12 hours in advanced (#6) was lower than the mean abs. error for SJ using forecasts made 48hrs in advance. This means that there were more accurate results when the observed_temp was compared to the forecast_temp using forecasts only made 12 hours in advance. This makes sense since meteorologists would most likely have greater success finding the actual temperature of a specific day the closer that they are to that date/time. Having a higher mean abs. error means there was more difference between the observed_temp and forecast_temp, which in hand means less successful predictions.

9. Pick two cities in this dataset. Investigate whether the forecast accuracy is better for one city than for the other, using an appropriate statistic. Discuss your findings.

```
library(dplyr)
alb_ot_ft_24 <- weather_forecasts |>
  filter(city == "ALBANY", forecast_hours_before == 24) |>
  mutate(abs_error_24 = abs(observed_temp - forecast_temp))

mean(alb_ot_ft_24$abs_error_24, na.rm = T)
```

```
[1] 2.277523
```

```
cle_ot_ft_24 <- weather_forecasts |>
  filter(city == "CLEVELAND", forecast_hours_before == 24) |>
  mutate(abs_error_24 = abs(observed_temp - forecast_temp))

mean(cle_ot_ft_24$abs_error_24, na.rm = T)
```

```
[1] 2.107798
```

Using the code snippet above, I found the mean abs. error between the observed_temp and the forecast_temp between the cities of Cleveland (Ohio) and Albany (NY). The first number 2.277523 displayed is the mean abs. error for Albany. The second number 2.107798 is the mean abs. error for Cleveland. The average difference between the observed_temp and the forecast_temp was lower for the city of Cleveland, so we can conclude that that the forecast accuracy is more accurate in Cleveland than the city of Albany.

B. Find your own data

For this component, pick a [Tidy Tuesday dataset]() and complete the following activity.

10. Provide a brief description of your dataset. Identify at least two questions you could try to answer using this dataset.

The dataset that I chose represents the sales data for the Diwali festival. It gives data such as age, gender, location, occupation of people who purchased items from retail stores during the Diwali festival period in India. The first question that comes to mind is what is the average amount of a transaction during the Diwali festival in India. Another question that can be answered is what regions/cities spent the most on supplies for the festival.

11. Open your dataset in R and compute one or more descriptive statistics that shed light on your questions. Discuss your findings.

```
setwd("C:/Users/bdish/OneDrive/Documents/Math167R/data")
diwali_sales <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/ma
write.csv(diwali_sales, "diwali_sales_data.csv")
```

```
sales_column <- diwali_sales$Amount
mean(sales_column, na.rm = T)
```

```
[1] 9453.611
```

```
diwali_sales |>
  group_by(State) |>
  summarize(top_state_spender = sum(Amount, na.rm = TRUE)) |>
  ungroup() |> arrange(desc(top_state_spender))
```

```
# A tibble: 16 × 2
   State            top_state_spender
   <chr>                        <dbl>
 1 Uttar Pradesh            19374968
 2 Maharashtra              14427543
 3 Karnataka                13523540
 4 Delhi                    11603819.
 5 Madhya Pradesh            8101142
 6 Andhra Pradesh            8037147.
 7 Himachal Pradesh          4963368
 8 Haryana                   4220175
 9 Bihar                     4022757
10 Gujarat                   3946082
11 Kerala                    3894492.
12 Jharkhand                 3026456
13 Uttarakhand               2520944
14 Rajasthan                 1909409
15 Punjab                    1525800
16 Telangana                 1151490
```

Based on the following code snippets, we can see that after loading the dataset, in the second code snippet I write the code to find the average price of a transaction from the amount column. We can see that the average amount spent on each transaction was roughly 9453 (assuming currency is Rupee). In the 3rd code segment, I am simply grouping the dataset by the Sales column, and then summarizing a variable which sums the total amount spent on each transaction based on a particular state. Simly, it's just summing the total amount spent per state. The final part of the code rearranges the data to assorted in descending order, so we can see what states spent the most first.

12. Are there any limitations of your analysis? Could additional data or more complicated methods improve your analysis? Discuss.

The limitations of my analysis include outliers, as the amount spent across different categories can vary significantly. Another issue that could arise from my analysis is the lack of factors that I did not include which can influence the result of the data. I believe that this dataset has enough data to make conclusions based on factors such as demographics, name, age group and the use of more complicated methods involving these factors/influences would help improve my analysis. It would provide a deeper insight into total spending across different age groups and regions, along with the categories where most/least spending occurred.